

Better approximation to the susie-love model

1 Model discussion

Last time we considered the susie-relax model, where the uncertainty of R only affects the mean, not the variance:

$$\begin{cases} x|R, \beta, \gamma \sim \mathcal{N}(\beta R_\gamma, \bar{R}/N) \\ R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1). \end{cases} \quad (1)$$

which leads to the conditional distribution:

$$p(x_{-j} | x_j, \beta, \gamma = 1) = \mathcal{N}\left(x_j \bar{R}_{j,-j}, \left(\frac{1}{N} + \frac{\beta^2}{\nu_0 + 3}\right) \bar{S}_j\right), \quad (2)$$

where $\bar{S}_j$ is the Schur complement of $\bar{R}_{jj}$ in $\bar{R}$. Because this depends on β , this model does not satisfy the “irrelevance of null SNPs” property. Is there another model that arises from the conditional distribution:

$$p(x_{-j} | x_j, \beta, \gamma = j) = \mathcal{N}\left(x_j \bar{R}_{j,-j}, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0 + 3}\right) S_j\right)? \quad (3)$$

2 Some new models with better approximation to susie-love

Recall our original SER model from ‘susie-love’:

$$\begin{cases} x|R, \beta, \gamma \sim \mathcal{N}(\beta R_\gamma, R/N) \\ R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1). \end{cases} \quad (4)$$

In the following, we show that it satisfies “irrelevance of the null SNPs” and, in fact, leads to conditional distribution (3). The novelty here is what we learned from susie-relax last week: To represent R in terms of blocks.

2.1 Model A (equivalent to the original SER-love model)

Let’s write down $p(x|R, \beta, \gamma = 1)$, other cases of γ are similar. Let

$$d_1 = R_{11} \in \mathbb{R}_+, \quad z_1 = \frac{R_{-1,1}}{d_1} \in \mathbb{R}^{J-1},$$

and

$$S_1 = R_{-1,-1} - d_1 z_1 z_1^\top$$

be the Schur's complement of R_{11} in R . Hence, the latent R is represented as

$$R = \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}. \quad (5)$$

Similarly, denote

$$\bar{S}_1 = \bar{R}_{-1,-1} - \bar{R}_{-1,1} \bar{R}_{-1,1}^\top$$

be the Schur's complement of $\bar{R}_{11}$ in $\bar{R}$, and $\mu_1 = \bar{R}_{-1,1}$, we can write

$$\bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix}. \quad (6)$$

Properties of IW distribution (together with the fact that $\bar{R}_{11} = 1$) imply:

$$\begin{cases} d_1 \sim \text{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\bar{R}_{-1,1}, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \text{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \end{cases} \quad (7)$$

The original SER version of Susie-love (with $\gamma = 1$; similar for other γ) becomes:

$$\begin{cases} x | \beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}\right) \\ d_1 \sim \text{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \text{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \quad (8)$$

Write $x = \begin{pmatrix} x_1 \\ x_{-1} \end{pmatrix}$. From the block normal distribution in (8), we have

$$p(x | \gamma = 1, \beta, d_1, z_1, S_1) = \mathcal{N}\left(x_1 | \beta d_1, \frac{d_1}{N}\right) \mathcal{N}_{J-1}\left(x_{-1} | x_1 z_1, \frac{S_1}{N}\right).$$

The key simplification is that, **after conditioning on x_1 , the conditional mean of x_{-1} is $x_1 z_1$ and no longer contains β or d_1** (relevant to our earlier discussion). Now we proceed to show that this satisfies the “irrelevance of null SNPs” property:

1. Marginalize out $z_1 | S_1$ leads to:

$$p(x | \gamma = 1, \beta, d_1, S_1) = \mathcal{N}\left(x_1 | \beta d_1, \frac{d_1}{N}\right) \mathcal{N}_{J-1}\left(x_{-1} | x_1 \mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) S_1\right).$$

What I found so interesting is that the conditional distribution $p(x_{-1} | x_1, S_1, \gamma = 1)$ resembles what was suggested by Matthew last time (cf. (3)).

2. Marginalize over S_1 . Thus

$$p(x \mid \gamma = 1, \beta, d_1) = \mathcal{N}\left(x_1 \mid \beta d_1, \frac{d_1}{N}\right) t_{J-1, \nu_0+3}\left(x_{-1} \mid x_1 \mu_1, \frac{\nu_0}{\nu_0+3} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) \bar{S}_1\right),$$

where the second argument is the multivariate t scale matrix and is denoted by $T_1(x)$:

$$T_1(x) \propto |\bar{S}_1|^{-1/2} \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right)^{-(J-1)/2} \left[1 + \frac{(x_{-1} - x_1 \mu_1)^\top (\nu_0 \bar{S}_1)^{-1} (x_{-1} - x_1 \mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}}\right]^{-(\nu_0+J+2)/2}.$$

Using the block inverse formula (notice the block form of $\bar{R}$ (6)),

$$x^\top \bar{R}^{-1} x = x_1^2 + (x_{-1} - x_1 \mu_1)^\top \bar{S}_1^{-1} (x_{-1} - x_1 \mu_1),$$

so

$$1 + \frac{(x_{-1} - x_1 \mu_1)^\top (\nu_0 \bar{S}_1)^{-1} (x_{-1} - x_1 \mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}} = \frac{x^\top \bar{R}^{-1} x + \nu_0/N}{x_1^2 + \nu_0/N}.$$

Therefore

$$T_1(x) \propto |\bar{S}_1|^{-1/2} \left(x_1^2 + \frac{\nu_0}{N}\right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1} x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2}. \quad (9)$$

When comparing different values of γ , **the final factor is common across SNPs**. Also, since $\bar{R}_{11} = 1$, $|\bar{R}| = |\bar{S}_1|$; analogously, $|\bar{S}_j| = |\bar{R}|$ for any SNP j with $\bar{R}_{jj} = 1$.

3. Marginalize over β and d_1 finally leads to

$$p(x \mid \gamma = 1) = T_1(x) \int_0^\infty \mathcal{N}\left(x_1 \mid 0, \frac{d_1}{N} + d_1^2 \sigma_0^2\right) d\text{IG}\left(d_1; \frac{\nu_0+2}{2}, \frac{\nu_0}{2}\right).$$

This is a one-dimensional integral over $d_1 > 0$ which does not have closed-form expression. Cancel out the first (determinant term) and the last (quadratic term) in $T_1(x)$ because they are common for all SNPs, we have

$$p(\gamma = j \mid x) \propto \left(1 + \frac{N x_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \int_0^\infty \mathcal{N}\left(x_j \mid 0, \frac{d_j}{N} + d_j^2 \sigma_0^2\right) d\text{IG}(d_j). \quad (10)$$

Conclusion. The resulting PIP weight depends on SNP j only through x_j .

2.2 Model B

Simplify $d_1 \equiv 1$ to avoid intractable integral (and we know the diagonal of the in-sample LD matrix is 1 anyway), we have the following model (model B):

$$\begin{cases} x \mid \beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix}, \frac{1}{N} \begin{pmatrix} 1 & z_1^\top \\ z_1 & S_1 + z_1 z_1^\top \end{pmatrix}\right) \\ z_1 \mid S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \quad (11)$$

This model leads to

$$p(x \mid \gamma = 1) \propto |\bar{R}|^{-1/2} \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1}x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2} \mathcal{N}\left(x_1 \mid 0, \frac{1}{N} + \sigma_0^2\right).$$

Since $|\bar{R}|$ and $x^\top \bar{R}^{-1}x$ do not depend on which SNP is chosen, the PIP weights simplify further. Dropping the common factors gives

$$p(\gamma = j \mid x) \propto \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \mathcal{N}\left(x_j \mid 0, \frac{1}{N} + \sigma_0^2\right). \quad (12)$$

Compare to SER-RSS's PIP. Recall SER-RSS's PIP:

$$\tilde{p}(\gamma = j \mid x) \propto \exp\left(\frac{N\sigma_0^2}{N\sigma_0^2 + 1} \frac{Nx_j^2}{2}\right) \quad (13)$$

We now show that (12) tends to (13) as $\nu_0 \rightarrow \infty$. Since

$$\left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} = \left(1 + \frac{2}{\nu_0 + 3} \times \frac{\nu_0 + 3}{\nu_0} \frac{Nx_j^2}{2}\right)^{(\nu_0+3)/2} \asymp \exp\left\{\frac{\nu_0 + 3}{\nu_0} \frac{Nx_j^2}{2}\right\}$$

Also,

$$\mathcal{N}\left(x_j \mid 0, \frac{1}{N} + \sigma_0^2\right) \propto \exp\left\{-\frac{x_j^2}{2(1/N + \sigma_0^2)}\right\} = \exp\left\{-\frac{1}{1 + N\sigma_0^2} \frac{Nx_j^2}{2}\right\}.$$

Therefore the limiting weight in (12) is asymptotically (as ν_0 gets large)

$$\exp\left\{\left(\frac{\nu_0 + 3}{\nu_0} - \frac{1}{N\sigma_0^2 + 1}\right) \frac{Nx_j^2}{2}\right\} = \exp\left\{\left(\frac{N\sigma_0^2}{N\sigma_0^2 + 1} - \frac{1}{\nu_0 + 3}\right) \frac{Nx_j^2}{2}\right\} \approx \exp\left\{\frac{N\sigma_0^2}{N\sigma_0^2 + 1} \frac{Nx_j^2}{2}\right\}$$

which is exactly the SER-RSS PIP weight in (13). Hence, for finite ν_0 , PIP under SER-love model (B) is slightly more diffused than susie-rss.

Inference under model B (11). For a general SNP j , write

$$\mu_j = \bar{R}_{-j,j}, \quad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top, \quad \kappa_j = \nu_0 + Nx_j^2.$$

Under model B, conditional on $\gamma = j$, the likelihood factorizes as

$$p(x \mid \beta, z_j, S_j, \gamma = j) = \mathcal{N}\left(x_j \mid \beta, \frac{1}{N}\right) \mathcal{N}_{J-1}\left(x_{-j} \mid x_j z_j, \frac{S_j}{N}\right).$$

Thus β is conditionally independent of (z_j, S_j) given x and $\gamma = j$, and

$$p(\beta, z_j, S_j \mid x, \gamma = j) = p(\beta \mid x_j, \gamma = j) p(z_j, S_j \mid x, \gamma = j).$$

- The posterior for β is the usual normal-normal update:

$$\beta \mid x, \gamma = j \sim \mathcal{N}\left(\frac{N}{N + \sigma_0^{-2}}x_j, \frac{1}{N + \sigma_0^{-2}}\right).$$

- For the LD-column variables, first condition on S_j . Combining

$$z_j \mid S_j \sim \mathcal{N}_{J-1}\left(\mu_j, \frac{S_j}{\nu_0}\right), \quad x_{-j} \mid x_j, z_j, S_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j z_j, \frac{S_j}{N}\right)$$

gives

$$z_j \mid S_j, x, \gamma = j \sim \mathcal{N}_{J-1}\left(\frac{\nu_0 \mu_j + N x_j x_{-j}}{\nu_0 + N x_j^2}, \frac{S_j}{\nu_0 + N x_j^2}\right).$$

- Integrating out z_j gives

$$x_{-j} \mid x_j, S_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j \mu_j, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right) S_j\right).$$

Therefore the posterior for S_j is inverse-Wishart:

$$S_j \mid x, \gamma = j \sim \mathcal{IW}\left(\nu_0 \bar{S}_j + \frac{N \nu_0}{\nu_0 + N x_j^2} (x_{-j} - x_j \mu_j)(x_{-j} - x_j \mu_j)^\top, \nu_0 + J + 2\right).$$

Equivalently, the joint posterior of (z_j, S_j) can be represented by first drawing S_j from this inverse-Wishart distribution and then drawing z_j from the conditional normal distribution above. Finally, the full posterior over γ uses the PIP weights in (12), normalized over $j = 1, \dots, J$.